ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PRACTICE EXAM 3

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 7 year: the design process, testing the strength of paper beams, forces in bridges, properties of materials, recycling and processing, levers in tools and the human body, linkages, circuit drawing, electromagnets and the electric bell.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings. Drawings must be neat and labelled.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which of these is a natural structure?

A Tree B Bridge C Chair D Tent

1.2 A twisting force is called …

A tension. B torsion. C compression. D shear.

1.3 Corrugated iron sheets are stronger than flat sheets because they are …

A painted. B heavier. C thinner. D folded.

1.4 A bottle opener is an example of a …

A second-class lever. B first-class lever. C third-class lever. D linkage.

1.5 A load of 900 N is lifted with an effort of 300 N. The mechanical advantage is …

A $\frac{1}{3}$ B 600 C 3 D 1 200

1.6 The SI unit of force is the …

A kilogram. B metre. C newton. D joule.

1.7 A circuit with a gap in it is called …

A an open circuit. B a closed circuit.
C a short circuit. D a parallel circuit.

1.8 The component that changes chemical energy into electrical energy is a …

A bulb. B switch. C motor. D cell.

1.9 The step of the design process in which you test and judge your product is …

A investigate. B evaluate. C design. D make.

1.10 A drawing that shows an object as seen from directly above is the …

A front view. B side view. C isometric view. D plan (top) view.

[10]

QUESTION 2: STRUCTURES

2.1 Give ONE example of a natural frame structure and ONE example of a manufactured shell structure. (2)

2.2 Grade 7 learners tested beams made from identical sheets of paper. Each beam was placed across a 20 cm gap and masses were added until it collapsed.

Shape of paper beam	Flat sheet	Folded into a U-shape	Rolled into a tube
Largest mass held (g)	50	400	650

2.2.1 Which beam was the strongest? (1)

2.2.2 Explain why folding or rolling the paper makes it stronger. (2)

2.2.3 Name TWO things that had to be kept the same to make this a fair test. (2)

2.2.4 Name ONE type of graph that could be used to show these results. (1)

2.3 When a heavy truck drives over a beam bridge, the beam bends slightly. Which part of the beam is in compression, and which part is in tension? (2)

2.4 Explain why the frame of a bicycle is made of hollow metal tubes. (2)

2.5 Name the FIVE steps of the design process in the correct order. (5)

2.6 What is meant by the *aesthetics* of a product? (1)

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QUESTION 3: MATERIALS AND PROCESSING

3.1 Explain what is meant by each property: (a) flexibility; (b) hardness. (2)

3.2 Suggest a suitable material for the top of a school desk and give a reason for your choice. (2)

3.3 Explain what is meant by *reduce*, *reuse* and *recycle*, giving an example of each. (3)

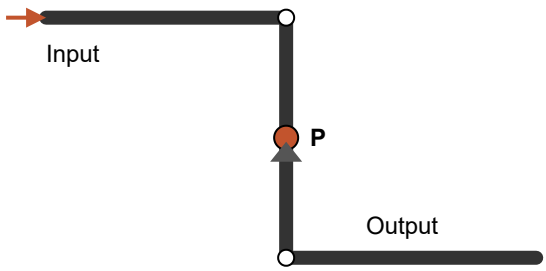
3.4 Describe, in THREE steps, how paper is made from trees. (3)

3.5 Why is it important to use materials that are sustainable? (2)

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QUESTION 4: MECHANICAL SYSTEMS

4.1 Study the linkage below. Point P is a fixed pivot.



4.1.1 Name this type of linkage. (1)

4.1.2 In which direction does the output rod move when the input rod is pushed to the right? (1)

4.1.3 What would happen to the output movement if pivot P were moved closer to the input end? (1)

4.2 A crowbar (first-class lever) is used to lift a rock with a weight of 1 200 N. The effort arm is 1,5 m and the load arm is 0,5 m.

4.2.1 Calculate the mechanical advantage. Use: MA = effort arm ÷ load arm. (2)

4.2.2 Calculate the effort needed to lift the rock. (2)

4.2.3 The fulcrum is moved so that the load arm is 0,3 m and the effort arm is 1,7 m. Calculate the new mechanical advantage (round off to two decimal places) and say what happens to the effort needed. (3)

4.3 Name the class of lever: (a) your forearm when you lift a cup; (b) your foot when you stand on your toes. (2)

4.4 Explain what *mechanical advantage* means in your own words. (2)

4.5 Give TWO ways to reduce friction in a mechanism. (2)

4.6 Give TWO examples of linkages in everyday life. (2)

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QUESTION 5: ELECTRICAL SYSTEMS

5.1 Draw a neat circuit diagram with ONE cell, ONE switch, ONE bulb and ONE buzzer connected in a single loop. Use the correct symbols. (4)

5.2 Learners wound different numbers of coils of wire around an iron nail connected to the same battery and counted how many paper clips the electromagnet could pick up.

Number of coils	10	20	30	40
Number of paper clips picked up	3	7	10	14

5.2.1 Describe the relationship between the number of coils and the strength of the electromagnet. (2)

5.2.2 Predict how many paper clips 50 coils would pick up. (1)

5.2.3 Name the dependent variable in this investigation. (1)

5.2.4 Give ONE safety precaution for this investigation. (1)

5.3 Explain how an electric bell uses an electromagnet to make a ringing sound. (3)

5.4 Name a device that produces electricity from: (a) movement; (b) sunlight. (2)

5.5 What is a short circuit, and why is it dangerous? (2)

5.6 What do the letters LED stand for? (1)

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TOTAL: 75

END OF QUESTION PAPER